

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Semester :VII

**Branch : Electronics and Communication Engineering/Electronics and
Telecommunication Engineering**

Subject Code & Name: BTETPE702D and Fiber Optic Communication

Max Marks: 60

Date:04-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) A typical relative refractive index difference for an optical fiber designed for longdistance transmission is 1%. Estimate the NA and the solid acceptance angle in air for the fiber when the core index is 1.46. Further, calculate the critical angle at the core-cladding interface within the fiber. It may be assumed that the concepts of geometric optics hold for the fiber.	CO1/CO4	6
B) Explain the skew ray propagation in an optical fiber. Derive the acceptance conditions for skew rays.	CO1	6
C) List the applications of optical fiber communication.	CO1	6
Q.2 Solve Any Two of the following.		12
A) A multimode step index fiber with a core diameter of 80μm and a relative index difference of 1.5% is operating at a wavelength of 0.85μm. If the core refractive index is 1.48, estimate: (a) the normalized frequency for the fiber; (b) the number of guided modes.	CO2/CO4	6
B) Explain the different types of optical fibers with a neat diagram.	CO2	6
C) Define dispersion. Explain in detail with a schematic diagram	CO2/CO4	6
Q.3 Solve Any Two of the following.		12
A) With a schematic diagram explain the semiconductor injection LASER.	CO3	6
B) Derive the expression for LED power and efficiency.	CO3/CO4	6
C) Explain the operation of PIN diode.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain Wavelength division multiplexing	CO4/CO5	6

B)	Explain the EDFA amplifier.	CO5	6
C)	Explain Raman amplifier.	CO5	6
Q. 5 Solve Any Two of the following.			12
A)	Explain the concept of self-phase modulation.	CO5	6
B)	Explain group velocity dispersion.	CO4/CO5	6
C)	Explain the nonlinear effects in fiber optic links	CO5	6

***** End *****